MACHINE LEARNING CANVAS | Plant Disease Diagnosis — Phytosanitary Decision-Support System

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Methodology: OwnML (Louis Dorard Framework) | **Milestone:** M1 Inception Approved

Benchmark: PlantVillage (54,306 images, 38 classes) | SLA: Latency $p95 < 200\,\text{ms}$, E2E $< 5\,\text{s}$ | Version: 1.0 (Oct 2026)

1. VALUE PROPOSITION

Core Objective: Turn a single leaf photo into an immediate, actionable phytosanitary diagnosis within seconds, preventing catastrophic crop loss for underserved farmers.

- **Global Problem Impact:** Crop diseases destroy **20–40% global yield** annually (FAO), generating >**$220B** in damages. Smallholders lack agronomists, and lab cultures take weeks, causing crop death and blind pesticide overuse.
- **Value Equation (Before → After):**
  - → *Turnaround:* Days/weeks (lab culture/PCR) → < 5 **seconds** in-field.
  - → *Cost per test:* $10–50 (consultant/travel) → ∼**$0.001** cloud inference.
  - → *Efficacy:* Calendar chemical spraying → **Targeted, organic-first advisory**.
  - → *Access:* 1 expert : 1,000 farmers → **1 model : millions via smartphone**.
- **Target Personas & Core Workflows:**
  - **Smallholder Farmer** (0.5–5 ha, Kenya/India/Brazil): Photo → diagnosis + organic protocol in local language (**UC-01**, offline queue **UC-02**).
  - **Extension Officer**: Batch upload 20+ farm photos → prioritized triage (**UC-03**).
  - **Hobby / Urban Gardener**: Identifies benign spots → stops chemical waste.
  - **Agri-Tech Platform / Co-op**: White-label diagnostic engine via REST/MCP (**UC-04**).
- **System Interfaces:** Mobile PWA, WhatsApp/Telegram bot (MCP tool), IoT Gateway.

2. DATA SOURCES

Benchmark & Real-World Data Repositories:

- **Primary Benchmark: PlantVillage (Mohanty et al., 2016):**
  - **Volume:** 54,306 high-resolution leaf photographs, expert-verified.
  - **Taxonomy:** 38 discrete classes across 14 crop species: *Apple, Blueberry, Cherry, Corn, Grape, Orange, Peach, Bell Pepper, Potato, Raspberry, Soybean, Squash, Strawberry, Tomato*.
  - **Pathology Coverage:** 26 infectious conditions (fungal blights, rusts, bacterial spots, viral mosaics) + 12 healthy control classes.
  - **Source Registry:** Hugging Face datasets (`mohanty/PlantVillage`).
- **Production Real-World Ingestion Streams:**
  - Farmer field captures uploaded via mobile app and messaging bots.
  - Agricultural extension tablet batch uploads with field validation metadata.
  - IoT greenhouse camera feeds for continuous anomaly scouting (**UC-05**).
- **Phytosanitary Knowledge Bases (Treatment Mappings):**
  - Structured domain knowledge from FAO and CABI Crop Protection Compendium.
  - Mapping: `(Crop, Pathology)` → `(Cultural, Organic, Chemical, Prevention)`.
- **Future Contextual Priors (v2.0 Roadmap):**
  - Agro-climatic weather APIs (temperature, humidity) for Bayesian prior likelihoods.

3. DATA COLLECTION & GOVERNANCE

Data Ingestion, Versioning & Feedback Loops:

- **Reproducible Versioning & Lineage (DVC):**
  - Raw images and curated train/val/test splits tracked with Data Version Control (**DVC**). Immutable hash pointers stored in Git; raw assets stored in cloud object storage (S3/GCS).
  - Pipelines locked via `dvc.yaml`; single-command reproduction via `dvc repro`.
- **Active Learning & Continuous Feedback Loop:**
  - **Uncertainty Sampling:** Predictions with entropy in $[0.50, 0.75]$ automatically routed to human agronomist verification queue.
  - **Dispute Ingestion:** Farmer feedback mechanism ("Dispute result" / "Report misdiagnosis") captures edge cases and seasonal disease variants.
- **Collection Constraints & Bottlenecks:**
  - Expert labeling cost: Certified phytopathologist review ∼$1.50/image.
  - Class imbalance: Prevalent classes (Tomato Yellow Leaf Curl: 5,357 images) vs rare classes (Potato Early Blight: 1,000 images). Addressed via focal loss / reweighting.
- **Privacy, Ethics & Regulatory Boundaries:**
  - **EXIF Stripping:** GPS coordinates discarded at ingestion; opt-in only for national disease tracking.
  - **Zero PII:** Strictly non-human biological imagery.
  - **Regulatory Disclaimer:** Certified as advisory support; not a legal pesticide mandate.

4. PREDICTION TASK (ML TASK)

Mathematical Problem Formulation:

- **Task Archetype:** Supervised Multi-Class Single-Label Image Classification over $K = 38$ mutually exclusive botanical-condition classes.
- **Input Entity:** Single leaf RGB image tensor $\mathbf{x} \in \mathbb{R}^{3\times224\times224}$ under natural ambient field lighting and varying backgrounds.
- **Output Target:** Calibrated posterior class distribution $\mathbf{p} = \text{Softmax}(\mathbf{z}/\tau) \in [0,1]^{38}$ ($\sum_{k=1}^{38} p_k = 1$), where $\mathbf{z} \in \mathbb{R}^{38}$ represents raw logits and $\tau > 0$ is a post-hoc temperature scaling parameter calibrated on the validation set.
- **Delivered Inference Payload:**
  - Top-k ranked predictions: $\left(\text{Crop}_i, \text{Condition}_i, \hat{p}_i\right)$ for $i \in \{1, \ldots, k\}$.
  - Calibrated confidence score $C = \max_k p_k$.
  - Out-of-Distribution (OOD) score $S_{\text{OOD}}(\mathbf{x})$ rejecting non-leaf/non-crop inputs.
  - Human-readable agronomic advisory (pathogen type, severity, cultural remedy).
- **Target Offline Acceptance Criteria:**
  - Top-1 Accuracy: $\geq 95.0\%$ | Macro-Averaged F1: ≥ 0.930.
  - Expected Calibration Error (ECE): < 0.050.

5. FEATURES & PREPROCESSING

Input Representations & Data Transformations:

- **Deterministic Preprocessing Pipeline:**
  - Format: 3-channel RGB image decode.
  - Geometry: Aspect-ratio preserving resize and center-crop to $224 \times 224 \times 3$.
  - Standardization: Zero-mean, unit-variance normalization with ImageNet statistics: $\boldsymbol{\mu} = [0.485, 0.456, 0.406]$, $\boldsymbol{\sigma} = [0.229, 0.224, 0.225]$.
- **Robustness Augmentations (Invariance Training):**
  - Spatial: Random rotation ($\pm35^\circ$), horizontal/vertical flips, affine shearing, perspective transforms (simulating non-planar field angles).
  - Photometric: Color jitter (brightness ±0.2, contrast ±0.2, saturation ±0.2, hue ±0.05), Gaussian blur (simulating smartphone defocus).
  - Occlusion: Random Erasing / CutOut (simulating soil splatter or leaf tears).
- **Deep Latent Representations:**
  - High-dimensional feature vector $\mathbf{f} \in \mathbb{R}^D$ ($D = 1280$ for EfficientNet, $D = 768$ for ViT) extracted prior to the final classification layer.
- **Inference Safeguards & Guardrails:**
  - Test-Time Augmentation (TTA, 5-crop averaging) on uncertain border cases.
  - Energy-based OOD detector on $\mathbf{f}$ to reject non-botanical uploaded images.

6. BUILDING & UPDATING MODELS

Model Architecture, MLOps Stack & Orchestration:

- **Candidate Architectures (Transfer Learning):**
  - *Production Backbone:* **EfficientNet-B0** (4.0M parameters, compound scaling, lowest latency, ideal for CPU/edge container deployment).
  - *Robust Residual Baseline:* **ResNet-50 / ConvNeXt-Tiny** (robust feature extractor).
  - *Vision Transformer:* **ViT-Base/16** (global multi-head self-attention).
- **Training Optimization Protocol:**
  - Loss Function: Cross-Entropy with Label Smoothing ($\epsilon = 0.1$) to prevent overconfident misdiagnoses.
  - Optimizer: AdamW (lr $= 3 \times 10^{-4}$, weight decay 10^{-4}); Cosine Annealing schedule.
- **Experiment Tracking & Registry:**
  - **MLflow**: Complete lineage tracking of hyperparameters, training curves, confusion matrices, and serialized model artifacts. Stage gates: *Candidate* → *Staging* → *Production*.
- **Pipeline Automation & Quality Gates:**
  - **Prefect DAGs**: Automated end-to-end retraining and evaluation workflows.
  - **Great Expectations**: Strict data contract validation prior to training.
  - **Deepchecks**: Automated model test suite verifying data leakage and performance.
  - **CI/CD**: GitHub Actions running linter (`Ruff`), unit tests (`pytest`), container build.

7. DECISIONS & OPERATIONAL ACTIONS

Translating Predictions into High-Impact Decisions:

- **Confidence-Gated Operational Policy:**
  - **High Confidence ($C \geq 0.80$):** *Action:* Direct automated diagnosis. Render primary pathology name, severity rating, and step-by-step cultural/organic intervention (e.g., copper fungicide, canopy pruning).
  - **Moderate Confidence ($0.60 \leq C < 0.80$):** *Action:* Top-2 differential diagnosis. Prompt user with a guided visual inspection checklist (e.g., *"Look at leaf underside: are concentric rings or powdery spores visible?"*).
  - **Low Confidence ($C < 0.60$) or OOD Flag:** *Action:* **Strict Abstention Policy**. System outputs *"Uncertain Diagnosis — please consult a local extension specialist or re-photograph with clear lighting"*. Prevents dangerous pesticide misuse.
- **Role-Specific Automated Decisions:**
  - *Farmer:* Instant plain-language advisory (English, Swahili, Hindi) delivered over WhatsApp or mobile app.
  - *Extension Officer:* Prioritized batch dashboard ranking farms by contagion severity for rapid field dispatch.
  - *Greenhouse IoT:* Automated activation of dehumidifiers / air scrubbers upon localized fungal detection.

8. MAKING PREDICTIONS (SERVING)

Runtime Serving Infrastructure & SLA Budgets:

- **Production Serving Modalities:**
  - **FastAPI REST Service:** Containerized microservice exposing `POST /predict` (multipart image upload) and `GET /health` with OpenAPI documentation.
  - **MCP Server Interface:** Model Context Protocol exposing the `diagnose_leaf` tool, allowing seamless native invocation by AI agents (Claude, ChatGPT, WhatsApp bot).
  - **Offline-First Mobile Sync:** Client SQLite queue buffering photos in zero-reception fields; background sync upon cell tower re-acquisition.
  - **Asynchronous Batch Worker:** Extension officer multi-image uploads processed asynchronously with worker queues.
- **Deployment Topology:**
  - Docker Compose architecture orchestrating API container, MLflow artifact store, and Prometheus metrics scraper on Cloud (AWS/GCP/Render).
- **Strict Latency Budget (SLA):**
  - Image Decode & Preprocessing: $\leq 40\,\text{ms}$.
  - Model Forward Pass: $\leq 90\,\text{ms}$ (CPU) / $\leq 15\,\text{ms}$ (GPU).
  - Postprocessing & Knowledge Lookup: $\leq 20\,\text{ms}$.
  - **Total Server Latency:** $p95 < 200\,\text{ms}$; **End-to-End User Turnaround:** $< 5\,\text{s}$.

9. OFFLINE EVALUATION & SIMULATION

Validation Protocol & Cost-Loss Matrix:

- **Validation Split Strategy:** Stratified 70% Train / 15% Validation / 15% Test split, preserved across all 38 classes with fixed random seed and locked via DVC.
- **Target Offline Metrics:**
  - Top-1 Accuracy: $\geq 95.0\%$ | Macro-Averaged F1: ≥ 0.930.
  - Expected Calibration Error (ECE): < 0.050 (ensuring probabilities represent true frequencies).
- **Asymmetric Cost-Utility Matrix (Impact Simulation):**
  - **True Positive (TP):** **High Positive Utility ($+100$)** → Crop saved, contagion halted.
  - **False Positive (FP)** (Healthy labeled as diseased): **Moderate Cost (-15)** → Farmer spends $5–15/ha on unnecessary organic spray; crop remains intact.
  - **False Negative (FN)** (Diseased labeled as healthy): **Catastrophic Loss (-100)** → Unchecked pathogen proliferation, total crop destruction (30–100% yield collapse).
  - **Pathogen Misidentification (CM):** **Severe Loss (-80)** → Ineffective treatment, phytotoxicity, delayed proper intervention.
- **Mitigation:** Threshold tuning minimizing FN for virulent classes + abstention fallback on uncertainty.

10. LIVE EVALUATION, TELEMETRY & DRIFT MONITORING

Business & Agronomic Impact KPIs:

- **Estimated Yield Saved:** Preserved harvest value computed via verified acreage diagnoses.
- **Chemical Reduction:** Tracking reduction in broad-spectrum prophylactic pesticide purchases.
- **Farmer Turnaround & SLA:** Median end-to-end diagnosis time $< 5\,\text{s}$; Weekly Active Farmers.
- **Expert Validation Agreement:** Expert agronomist audit on sampled cases ($\geq 92\%$ target).

Infrastructure Telemetry & API Monitoring:

- **Prometheus Exporter:** Auto-scraping `/metrics` every 15s.
- **Core Telemetry:** Throughput (RPS), HTTP 4xx/5xx error rates, latency percentiles ($p50, p95, p99$).
- **Resource Utilization:** CPU load, RAM allocation, GPU VRAM, worker queue depth.
- **Grafana Dashboards:** Live visualization of operational health, error budgets, and uptime.

Data Drift & Concept Drift Detection:

- **Alibi Detect / Evidently Engine:**
  - *Covariate Shift:* Kolmogorov-Smirnov test on low-level attributes (luminosity, contrast, blur).
  - *Embedding Drift:* Maximum Mean Discrepancy (MMD) on latent feature representations $\mathbf{f}$.
  - *Concept Drift:* Chi-squared test on predicted class distribution shifts across seasons.
- **Automated Alerting:** Threshold breach triggers Prefect automated retraining pipeline.